

EduGenie-Google Gemini Powered Learning Assistant
Government Arts and Science College , Manur – MSU142

Team Leader : Gayathri E

Team Members : Uma Maheswari S, Reka M, Nithersana M, Abinaya M

Brainstorming and Ideation Phase

1. Feature Expansion & Core Enhancements

- **Interactive Quiz Mode (UI Improvement)**
 - Convert raw JSON output into interactive HTML forms with radio buttons or cards.
 - Add immediate feedback (instant correct/incorrect highlight) and score calculation upon submission.
- **Smart Content Processing (Multi-Format Inputs)**
 - **File Uploads:** Allow users to upload PDFs, Word documents (.docx), or plain text files (.txt) using FastAPI's UploadFile and parse them into context automatically.
 - **URL Reader:** Add an input for webpage/article links, scraping main body text using libraries like BeautifulSoup or newspaper3k.
- **Flashcard Generator**
 - Generate key concept/definition pairs as flip cards using CSS animations, giving students a quick self-testing tool alongside quizzes.
- **Exporting & Note Management**
 - Allow users to download summaries, study paths, or generated quiz sheets as formatted PDF files or Markdown notes.

2. Technical & Architectural Roadmap

- **Async API Calls for Gemini**

- Refactor endpoint handlers and Gemini SDK calls to use async/await patterns or thread pools to prevent blocking the FastAPI event loop during model generation.
- **Structured Outputs & Schema Validation**
 - Use Pydantic schemas or Gemini's structured JSON output mode to strictly enforce response formats (e.g., ensuring quiz responses always adhere to a strict array schema without regex cleanup).
- **Client-Side Improvements (JS/AJAX)**
 - Replace static page reloads with lightweight asynchronous requests (fetch API) so users can generate quizzes, summaries, or Q&A responses dynamically without leaving the view.
- **API Key Security & Environment Handling**
 - Add startup checks in main.py to verify that GEMINI_API_KEY exists and is valid before accepting incoming requests.

3. UI/UX & Gamification Ideas

- **Difficulty Selector for Quizzes & Learning Paths**
 - Allow users to choose their preferred difficulty level (*Beginner*, *Intermediate*, *Advanced*) before generating content.
- **Progress & Study Session Tracker**
 - Use browser localStorage or SQLite/PostgreSQL database storage to track completed quizzes, view past summaries, and monitor study streaks.
- **AI Tutor Persona Tuning**

- Introduce tone controls for Q&A responses, such as *Socratic Teacher* (asks guiding questions), *ELI5* (Explain Like I'm 5), or *Exam Prep Coach* (concise, high-yield bullet points).